Chinatip Lawansuk（廖有福）

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Github: github.com/chinatip-l

Professional Summary

Driven by a passion for building robust systems from low level, I began my career as an Embedded System Engineer, where I developed firmware and infotainment systems for a fleet of over 1000 electric vehicles. To expand my knowledge from the system level to the chip level, I pursued a Master's degree in Electrical Engineering, focusing on digital IC design. This advanced study allowed me to dive deep into hardware design and optimisation, culminating in a thesis where I designed and implemented a CORDIC hardware accelerator with a custom caching system.

Expertises

Programming: Verilog, SystemVerilog, C, C++, Python, Assembly, Shell Script, TCL, Tensorflow
IC Design Software: Synopsys VCS, Synopsys Design Compiler (Design Vision), Synopsys TetraMAX, Cadence NC-Verilog, Cadence Innovus, Siemens Calibre
Skills: Cell-Based Digital IC Design Flow, Image Processing, Computer Architecture, Operating System, Embedded System Design, Firmware, RTOS, Circuit Design
Coursework: Computer-Aided VLSI System Design, Mixed-Signal IC Design, Computer Vision, Computer Architecture, Computer Organisation, Operating System

Research and Publication

CORDIC hardware Improvement using Content Addressable Memory Cache (Thesis)

- Investigated and designed an algorithm to cache partially calculated data and reuse it in the CORDIC architecture for sine and cosine calculation.
- Designed and implemented the processing core to suit the caching algorithm
- Designed and implemented a custom TCAM to use with the proposed cache unit and a complete cache subsystem from scratch, including multiple cache blocks, a PLRUm replacement policy, and a controller
- Designed and implemented a pipeline and synchronisation mechanism using a handshaking protocol
- Completed the full cell-based IC design flow from logic design and verification, synthesis, Design For Testability, Automatic Place and Route, post-layout simulation, and transistor-level simulation
- Using TSMC 90nm technology, resulting in a peak throughput of 67.796 MOPS, consumes 48.2 mW @ 338.98 MHz.

Two-Dimensional Convolution Accelerator of High-Precision Quantization (Conference Paper)

- Y. -T. Chao, J. -Y. Gao, **C. Lawansuk** and Y. -C. Fan, ”Two-Dimensional Convolution Accelerator of High-Precision Quantization,” *2024 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, NV, USA, 2024, pp. 1-2, doi: 10.1109/ICCE59016.2024.10444315.
- Developed a high-precision quantized 2D convolution accelerator leveraging the Winograd algorithm, achieving <0.2% quantization error.

Education

National Taipei University of Technology (國立臺北科技大學) Sept 2023 –Aug 2025
MSc in Electrical Engineering (GPA: 4.00/4.00)
- **Laboratory:** Integrated System Laboratory (ISLAB) @ NTUT

King Mongkut's Institute of Technology Ladkrabang (KMITL) Aug 2017 –Jul 2021
BE in Computer Engineering (GPA: 3.29/4.00)

Awards

First-Ranked Graduating Student (National Taipei University of Technology) May 2025
Second Class Honour (KMITL) Jul 2021

Experience

Firmware Engineer Intern, LinkCom Manufacturing Co., Ltd. – New Taipei, Taiwan Aug 2024 - May 2025
- Worked on a high-power-density DC-DC Converter including hardware design (Planar Transformer, LLC topology) and embedded software optimisation. Performance reaches more than 1 kW on the credit card size module.
- Initiate the project for the transformer test platform (Firmware, and Hardware)
- Investigate and design closed loop control system for power module and identify impact parameters

Embedded System Engineer, MuvMi Co., Ltd. – Bangkok, Thailand Oct 2021 - Sept 2023
- Develop IoT Fleet, Data Analysis and Visualisation Platform on AWS for EV Charging System, identified and mitigated up to 12% of energy loss
- Develop a Vehicle Control Unit Firmware (RTOS on arm Cortex M4 processor) and infotainment system (Buildroot on arm Cortex A7 processor) for Electric Minibus and Electric Tuk-Tuk for the fleet of 1000 vehicles

NATIONAL TAIPEI UNIVERSITY OF TECHNOLOGY

Taipei, Taiwan, R. O. C.

Name: Chinatip Lawansuk(Chinatip Lawa) Date of Birth: Feb. 16, 1999 Date Enrolled: Sep. 2023 Date Conferred: **** **** Date Issued: Feb. 20, 2025

Graduate Institute: International Program of Electrical Engineering and Computer Science(IEECS)

Degree Conferred: *** ***

The following transcript is hereby certified as correct according to the record of the university. Student ID: 112998405

Course	1st Semes. Credit	1st Semes. Grade	2nd Semes. Credit	2nd Semes. Grade
(2023 - 2024)				
Graduate Seminars	1.0#	94	1.0#	92
Detection and Estimation	3.0#	91		
Computer-Aided VLSI System Design and Practice	3.0#	92		
Operating Systems	3.0#	100		
Computer Vision	3.0#	90		
Deep Learning for Digital Image Analysis			3.0#	88
Error Control Technologies for Modern Mobile Communication Systems			3.0#	96
Mixed-Signal Integrated Circuit Design			3.0#	91
Data Science Principles with Applications on Educational Data			3.0#	87
Conduct	---	86	---	85
Total Credits & Average	13.0	93.3	13.0	90.6
(2024 - 2025)				
Master's Thesis	3.0#		3.0#	
Conduct	---	85	---	
Total Credits & Average	0.0	---		
Thesis				****
Total Credits "26.0"				
Total Average "91.96"				
GPA "4.00"				
Rank in Department : 1/8				
Total EMI Credits：26.0				
TO BE CONTINUED				

Remarks: The grading system is as follows: 80 or more=A; 70 to 79=B; 60 to 69=C; 50 to 59=D; less than 50=E; 70=the passing grade; W=Withdrawal; CW=Course Waived; P=Passing; F=Failing; #=English as a Medium of Instruction.

1912 NATIONAL TAIPEI UNIVERSITY OF TECHNOLOGY 教務專用章 國立臺北科技大學

Signature: Wan-Ting Huang
Registrar

Signature: Yuh-Shyan Hwang
Provost of Academic Affairs

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LISTENING AND READING TEST
OFFICIAL INSTITUTIONAL SCORE REPORT

Name CHINATIP LAWANSUK

Date of Birth FEBRUARY 16, 1999

ID Number 1100501503550

Test Date NOVEMBER 7, 2022

Client PERSONAL

LISTENING	TOTAL SCORE
460	875
READING	
415	

Report is valid for two years from the test administration date.

TOEIC® Services Thailand -- Certified Score Report -- Issued NOVEMBER 8, 2022

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